Lecture 17: Central Limit Theorem, Strong Law of Large Numbers

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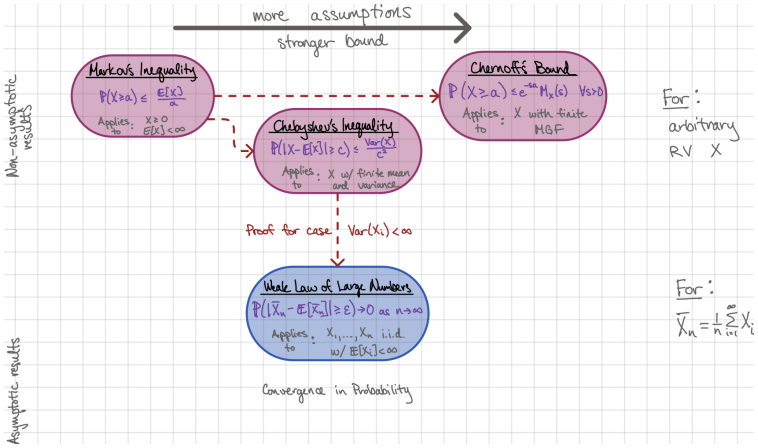
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0.1 Recap

Chernoff's bound requires the MGF to exist and be finite, which is stronger than assuming finite variance (Chebyshev) or mean (Markov), but yields a much stronger bound!



Now, we turn back to asymptotic convergence results for $\bar{X}_n$ and Z_n!

1 Central Limit Theorem

What is the shape of the distribution of $Z_n = \dfrac{S_n - n\mu}{\sigma\sqrt{n}}$?

Recall: $E[Z_n] = 0$ and $\text{Var}(Z_n) = 1$.

1.1 Central Limit Theorem

Theorem (CLT)

Let $X_1, X_2, ...$ be i.i.d. with mean μ and variance σ^2. Define $Z_n = \frac{X_1+\cdots+X_n-n\mu}{\sigma\sqrt{n}}$. Then the CDF of Z_n converges to the standard normal CDF $\Phi(z)$,

$$\lim_{n\to\infty} P(Z_n \le z) = \Phi(z)$$

for every z.

- Random effects are often the sum of a large number of small random factors (e.g., real-world noise), so this result has a wide array of applications!
- We don't need tedious PMFs/PDFs in this case, just the standard normal CDF Φ!

1.2 Central Limit Theorem

One way to prove this result is to show instead that the MGFs converge!

Proof sketch. Assume the MGF of X_i, M_{X_i}, exists near 0. Since the X_i are identically distributed, we have M_{X_i} is the same for all i, call this M_X. (Without loss of generality, we take $\mu = 0$, $\sigma = 1$.)

1. By independence and linearity, with some work, we can show that the MGF of Z_n is $[M_X(s/\sqrt{n})]^n$.
2. Since $E[X] = 0$ and $E[X^2] = 1$: the second order Taylor series expansion around $u = 0$ satisfies $M_X(u) = 1 + \frac{u^2}{2} + o(u^2)$.
3. So $[M_X(s/\sqrt{n})]^n = \left[1 + \frac{s^2}{2n} + o\left(\frac{1}{n}\right)\right]^n \to e^{s^2/2}$, the MGF of $N(0,1)$.
4. MGF convergence implies convergence of the CDFs. □

1.3 Convergence in Distribution

Definition

The sequence of random variable Y_n **converges in distribution** to Y, $Y_n \to Y$, if $F_{Y_n}(y) \to F_Y(y)$ at every y where F_Y is continuous.

- The CLT says $Z_n \to N(0,1)$ in distribution.
- Convergence in probability $\Rightarrow$ convergence in distribution.
- Convergence in distribution $\not\Rightarrow$ convergence in probability.

1.4 Normal Approximation by CLT

Let $S_n = X_1 + \cdots + X_n$, where the X_i are i.i.d. random variables with mean μ and variance σ^2. If n is large, the probability $P(S_n \le c)$ can be approximated by treating S_n as if it were normal, according to the following process:

1. Calculate the mean $n\mu$ and the variance $n\sigma^2$ of S_n.
2. Calculate the normalized value $z = (c - n\mu)/\sigma\sqrt{n}$.

3. Use the approximation

$$P(S_n \le c) = P(Z_n \le z) \approx \Phi(z).$$

1.4.1 Example: polling, revisited

How many people must we poll so the estimate is within $\varepsilon = 0.01$ of p with probability at least 0.95?

We have $X_i \sim \text{Bernoulli}(p)$ i.i.d., so $\mu = p$ and $\sigma^2 = p(1-p) \le 1/4$. Since p is unknown, use the worst case $\sigma = 1/2$ (a larger σ means a larger required n).

Rewrite the event in terms of S_n.

$$P(|\bar{X}_n - p| \ge \varepsilon) = P(|S_n - np| \ge n\varepsilon) \approx 2P(S_n \le np - n\varepsilon),$$

where the approximation follows from symmetry of the standard normal distribution.

Step 1. The mean of S_n is $n\mu = np$ and its variance is $n\sigma^2$.

Step 2. Calculate the normalized value $z = (c - n\mu)/(\sigma\sqrt{n})$:

$c = n(p-\varepsilon)$: $\quad z = \dfrac{-n\varepsilon}{\sigma\sqrt{n}} = -\dfrac{\varepsilon\sqrt{n}}{\sigma}$.

Step 3. Use the approximation $P(S_n \le c) \approx \Phi(z)$:

$P(S_n \le n(p-\varepsilon)) \approx \Phi\left(-\dfrac{\varepsilon\sqrt{n}}{\sigma}\right) = 1 - \Phi\left(\dfrac{\varepsilon\sqrt{n}}{\sigma}\right)$, by the symmetry of Φ.

Adding the two tails,

$$P(|\bar{X}_n - p| \ge \varepsilon) \approx 2\left(1 - \Phi\left(\frac{\varepsilon\sqrt{n}}{\sigma}\right)\right).$$

Solve for n. We want this to be at most 0.05:

$$2\left(1 - \Phi\left(\frac{\varepsilon\sqrt{n}}{\sigma}\right)\right) \le 0.05 \iff \Phi\left(\frac{\varepsilon\sqrt{n}}{\sigma}\right) \ge 0.975.$$

From the normal table, $\Phi(1.96) \approx 0.975$, so

$$\frac{\varepsilon\sqrt{n}}{\sigma} \ge 1.96 \iff n \ge \left(\frac{1.96\,\sigma}{\varepsilon}\right)^2 = \left(\frac{1.96 \cdot 0.5}{0.01}\right)^2 = 98^2 = 9604,$$

where we have assumed that the Bernoulli RVs have the worst case variance, $\sigma^2 = 1/4$.

Unlike Chebyshev and Chernoff, this is an *approximation* (it treats S_n as exactly normal), not a guaranteed bound.

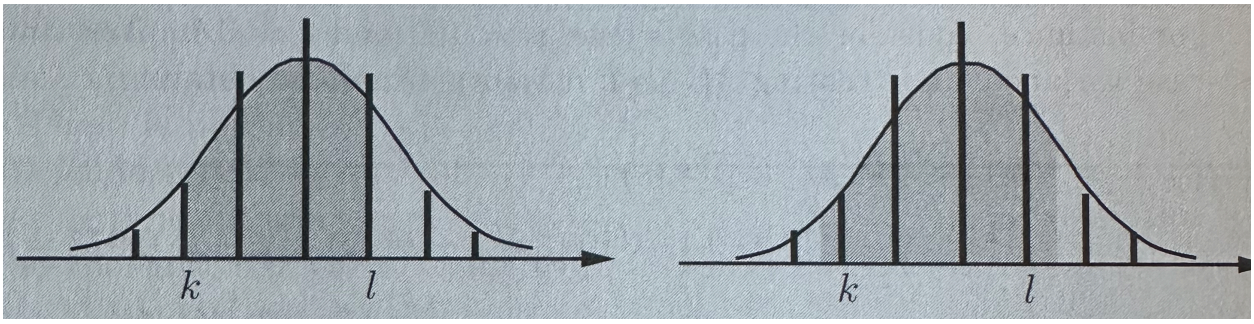
Method	$n \ge$
Chebyshev	50,000
Chernoff	18,445
CLT (approximation)	9,604

1.4.2 De Moivre–Laplace Approximation to Binomial

If $S_n \sim \text{Binomial}(n, p)$, then $\mu = np$, $\sigma^2 = np(1-p)$, and for large n, the previous process suggests that we should approximate

$$P(k \le S_n \le l) \approx \Phi\left(\frac{l - np}{\sqrt{np(1-p)}}\right) - \Phi\left(\frac{k - np}{\sqrt{np(1-p)}}\right).$$

However, when if we want to estimate $P(S_n = k)$ using $k = l$, this is a very bad estimate! Why?



De Moivre–Laplace approximation

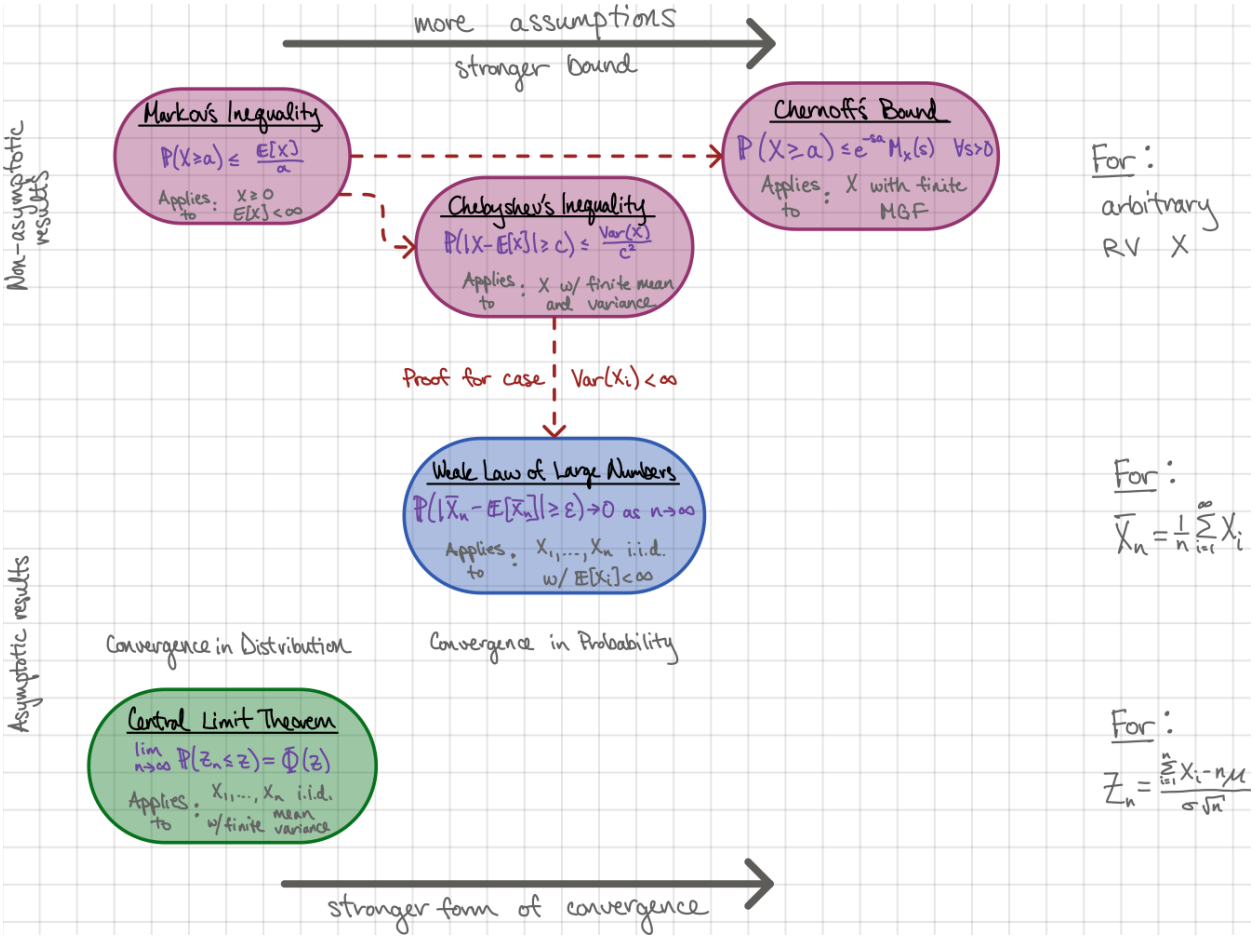
If $S_n \sim \text{Binomial}(n, p)$, n is large, and k and l are nonnegative integers, then

$$P(k \le S_n \le l) \approx \Phi\left(\frac{l + \frac{1}{2} - np}{\sqrt{np(1-p)}}\right) - \Phi\left(\frac{k - \frac{1}{2} - np}{\sqrt{np(1-p)}}\right).$$

The $\frac{1}{2}$ is called a *continuity correction*: it accounts for approximating a discrete distribution by a continuous one.

1.5 Summary

The CLT tells us that $Z_n \to N(0, 1)$ in distribution.



2 Strong Law of Large Numbers

Now, we turn back to $\bar{X}_n$ and seek to say something stronger than the WLLN.

The WLLN says that the probability of a significant deviation of $\bar{X}_n$ away from μ goes to 0 as $n \to \infty$. But, at each n, a bad event $\{|\bar{X}_n - \mu| \geq \varepsilon\}$ could happen as long as you run enough trials that you expect to see this very small probability event!

No matter how far right you move in the plot below, if you run many many trials, the WLLN does not guarantee that one of the lines will not pop out of the gray bar.

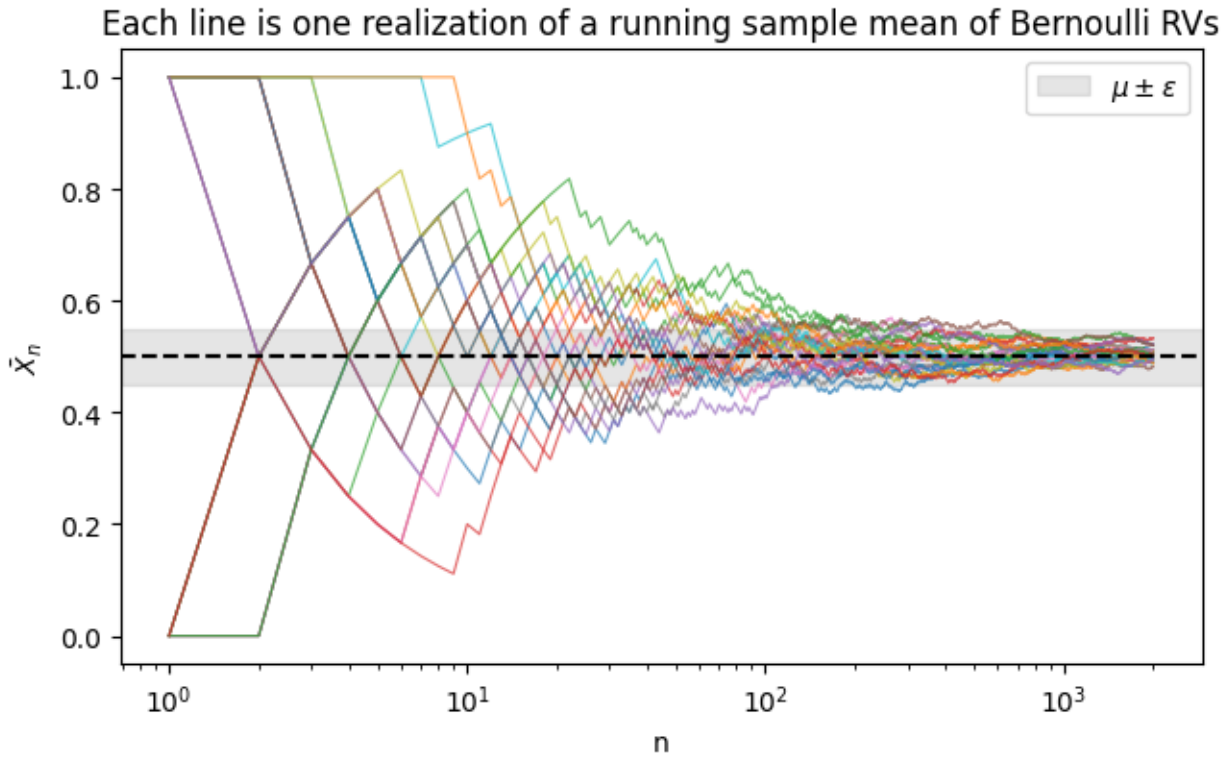
```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(151)

p, eps = 0.5, 0.05
paths, N = 25, 2000
ns = np.arange(1, N + 1)

X = rng.binomial(1, p, size=(paths, N))
xbar = np.cumsum(X, axis=1) / ns                    # row = one realization of X_1, ..., X_N
```

```
plt.figure(figsize=(6.5, 4.2))
for i in range(paths):
    plt.plot(ns, xbar[i], lw=0.8, alpha=0.7)
plt.axhline(p, color="k", ls="--")
plt.axhspan(p - eps, p + eps, color="gray", alpha=0.2, label=r"$\mu \pm \varepsilon$")
plt.xscale("log")
plt.xlabel("n")
plt.ylabel(r"$\bar{X}_n$")
plt.title("Each line is one realization of a running sample mean of Bernoulli RVs")
plt.legend()
plt.tight_layout()
plt.show()
```



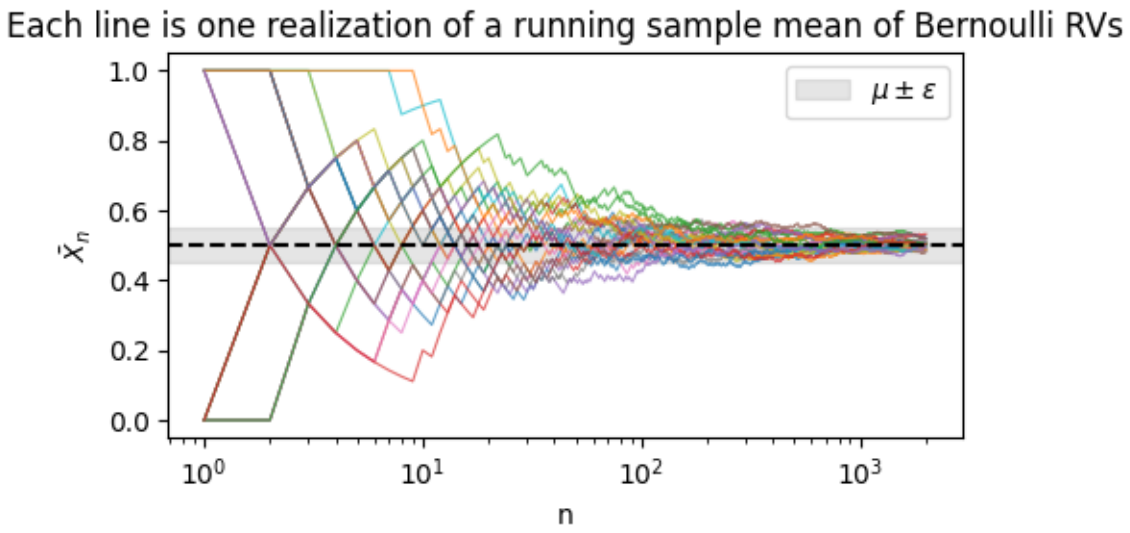
2.1 SLLN

Theorem (SLLN)

Let $X_1, X_2, \ldots$ be i.i.d. with mean μ. Then the sequence of sample means converges to μ with probability 1,

$$P\left(\lim_{n\to\infty} \bar{X}_n = \mu\right) = 1.$$

Interpretation. With probability 1, a single infinite running sample mean has the following property: for any $\epsilon > 0$, there is an n_0 (which depends on the run and on ϵ) such that for all $n \geq n_0$, the running sample mean stays within ϵ of μ, $|\bar{X}_n - \mu| < \epsilon$.



Relationship between SLLN and WLLN.

- SLLN $\Rightarrow$ WLLN.
- WLLN controls the set of $\bar{X}_n$ at each n separately; SLLN controls the whole sequence (for all n) at once.

2.2 Almost Sure Convergence

> **Definition**
>
> Let c be a real number. The sequence of random variables $Y_1, Y_2, \cdots$ **converges to** c **almost surely** if $P(\lim_{n\to\infty} Y_n = c) = 1$.

$$\text{almost sure conv.} \implies \text{conv. in probability} \implies \text{conv. in distribution.}$$

- converses are not true!
- SLLN: $\bar{X}_n \to \mu$ almost surely.

2.2.1 Example: minimum of uniform RVs

Let $U_1, U_2, \ldots$ be i.i.d. Uniform$(0,1)$ and $Y_n = \min(U_1, \ldots, U_n)$.

In any running trial, $Y_n \geq 0$ is monotonic nonincreasing with n and ≥ 0, so it must have a limit, $Y_n \to Y$.

For $0 < \epsilon \leq 1$, we have

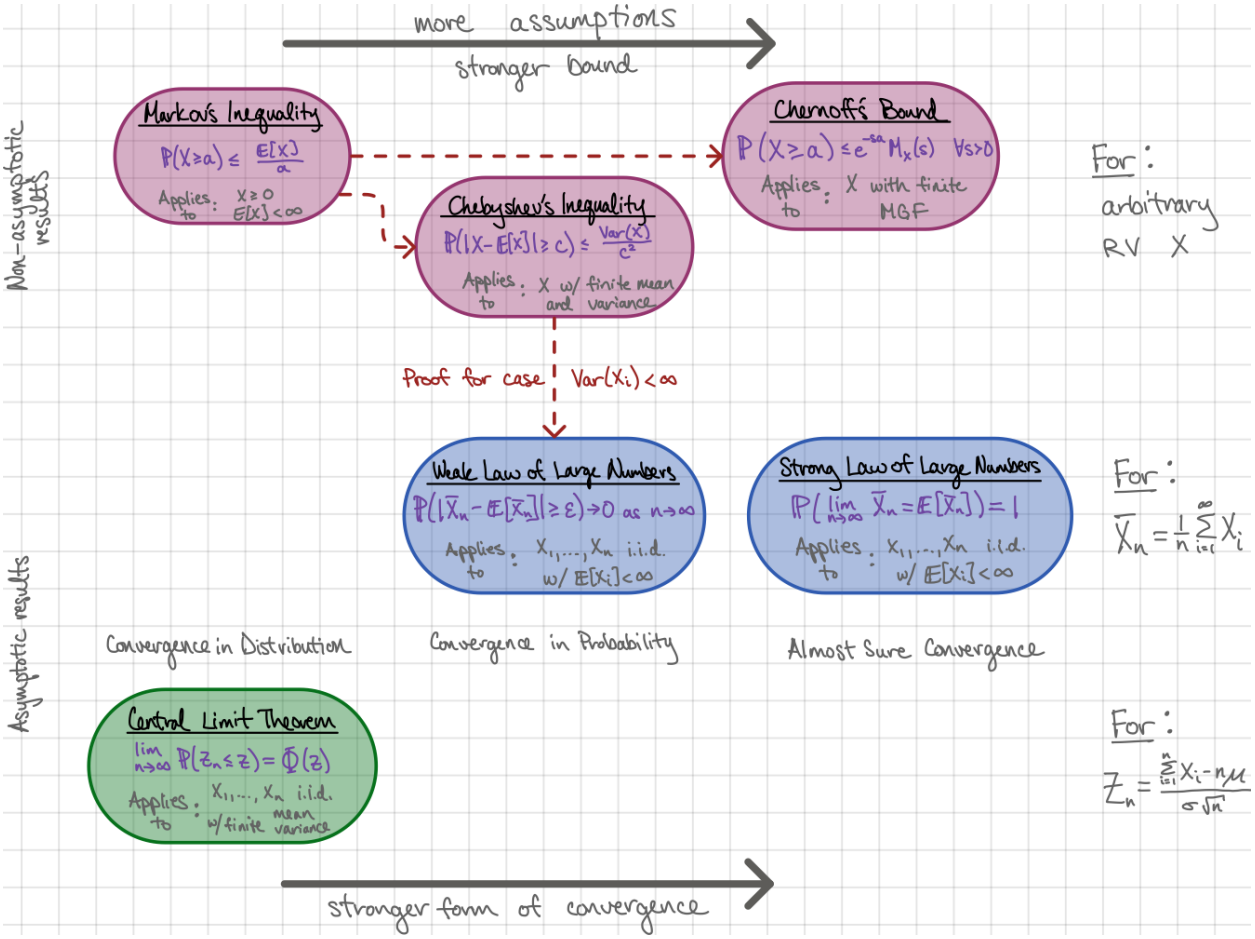
$$P(Y \geq \epsilon) \leq P(U_1 \leq \epsilon, \cdots, U_n \leq \epsilon) = (1-\epsilon)^n$$

for all n. Thus, $P(Y \geq \epsilon) \leq \lim_{n\to\infty}(1-\epsilon)^n = 0$.

Thus, $P(Y = 0) = 1$, and so Y_n converges to 0 *almost surely*!

2.3 Summary

The SLLN tells us that $\bar{X}_n \to \mu$ almost surely.


more assumptions
stronger bound
Non-asymptotic results
Markov's Inequality
$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$
Applies to: $X \geq 0$, $\mathbb{E}[X] < \infty$
Chernoff's Bound
$\mathbb{P}(X \geq a) \leq e^{-sa} M_X(s) \quad \forall s > 0$
Applies to: X with finite MGF
Chebyshev's Inequality
$\mathbb{P}(|X - \mathbb{E}[X]| \geq c) \leq \frac{\mathrm{Var}(X)}{c^2}$
Applies to: X w/ finite mean and variance
For: arbitrary RV X
Proof for case $\mathrm{Var}(X_i) < \infty$
Asymptotic results
Weak Law of Large Numbers
$\mathbb{P}(|\bar{X}_n - \mathbb{E}[\bar{X}_n]| \geq \varepsilon) \to 0$ as $n \to \infty$
Applies to: $X_1, \ldots, X_n$ i.i.d. w/ $\mathbb{E}[X_i] < \infty$
Strong Law of Large Numbers
$\mathbb{P}(\lim_{n\to\infty} \bar{X}_n = \mathbb{E}[\bar{X}_n]) = 1$
Applies to: $X_1, \ldots, X_n$ i.i.d. w/ $\mathbb{E}[X_i] < \infty$
For: $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{\infty} X_i$
Convergence in Distribution
Convergence in Probability
Almost Sure Convergence
Central Limit Theorem
$\lim_{n\to\infty} \mathbb{P}(Z_n \leq z) = \Phi(z)$
Applies to: $X_1, \ldots, X_n$ i.i.d. w/ finite mean variance
For: $Z_n = \frac{\sum_{i=1}^{n} X_i - n\mu}{\sigma\sqrt{n}}$
stronger form of convergence